FOOD ORDERING SYSTEM

Java Swing + JDBC Restaurant Point of Sale

PROJECT REPORT

Submitted by: _____

Roll Number: _____

Class / Semester: _____

Submitted to: _____

Institution: _____

1. Introduction

The Food Ordering System is a desktop restaurant POS application developed using Java Swing, JDBC and SQLite. It provides a graphical workflow for browsing products, managing carts, checking out, generating bills and reviewing orders.

2. Problem Statement

Manual order taking and bill preparation can lead to delays, calculation mistakes and poor record keeping. This application digitises the restaurant counter workflow and stores transactions locally.

3. Objectives

Create a multi-page GUI; display food products with category artwork; support search and filters; manage cart quantities; calculate subtotal, GST and total; persist orders using JDBC; save local receipts; provide order history and reports.

4. Main Features

Dashboard; product display cards; category filters and search; cart and quantity controls; customer details; Dine-in/Takeaway/Delivery; Cash/UPI/Card selection; 5% GST; receipt generation; order history; sales metrics and best sellers.

5. Technology Stack

Java 17, Java Swing, JDBC, SQLite, Maven and Java File I/O.

6. Java Concepts Demonstrated

Classes and objects; constructors; inheritance; polymorphism; abstraction; interfaces; encapsulation; collections; exception handling; GUI; JDBC; file handling; Swing event-dispatch threading.

7. System Design

FoodItem and its subclasses represent products. Cart stores selected items. MenuRepository and OrderRepository handle persistence. Database creates SQLite tables. Order captures a checkout snapshot. ReceiptService writes bills. POSFrame coordinates the interface.

8. Database Design

menu_items stores product details; orders stores order code, customer, payment, totals and timestamp; order_items stores each purchased item, quantity and unit price. order_items references orders through a foreign key.

9. Order Workflow

Start application → initialise SQLite → browse/search products → add products to cart → enter customer details → confirm order → persist order and line items → save TXT bill → refresh history and reports.

10. Testing Checklist

Verify menu loading, category filters, search, add/remove/update cart quantities, empty-cart validation, customer-name validation, GST calculations, successful order persistence, receipt creation, application restart and report refresh.

11. Limitations

Payment methods are demonstration labels only; no actual payment is processed. GST is set to 5%. Authentication, inventory, cloud synchronisation and printer integration are not included.

12. Future Enhancements

Add staff login, roles, menu editing, stock tracking, discounts, configurable taxes, PDF invoices, thermal printer support, charts, CSV exports and secure payment integration.

13. Conclusion

The project combines core Java programming concepts with a Swing POS interface, JDBC persistence, bill generation and local file handling to demonstrate a practical restaurant ordering workflow.